Vallalar Arts and Science College, Vadalur

Department of BCA

A Project Report on

CRICKET PLAYER PERFORMANCE TRACKING APPLICATION

Submitted by

J.Jancirani

M.Anupriya

M.Vinothini

S.Iswarya

BCA

Final Year

A Salesforce-Based Player Performance Analytics Platform

User Story

As a Salesforce Admin, I want to create a comprehensive player performance tracking application within Salesforce to capture and analyze detailed match-by-match information of cricket players. This application will include custom objects such as Player and Match Performance with carefully defined fields and relationships to ensure data integrity and accessibility. By leveraging master-detail and lookup relationships, the application will enable robust tracking of individual and team performances, facilitate insightful reporting and analytics, and support the overall management of player statistics, thereby enhancing strategic decision-making and player development initiatives.

1. Introduction

1.1 Project Overview

CRICKSTATS is a Salesforce-based application designed to track, manage, and analyze cricket players' performance on a match-by-match basis. The system uses custom objects, relationships, dashboards, and automation tools to provide structured performance insights.

It enables coaches, analysts, and administrators to record player statistics, generate reports, and support strategic decision-making.

1.2 Purpose

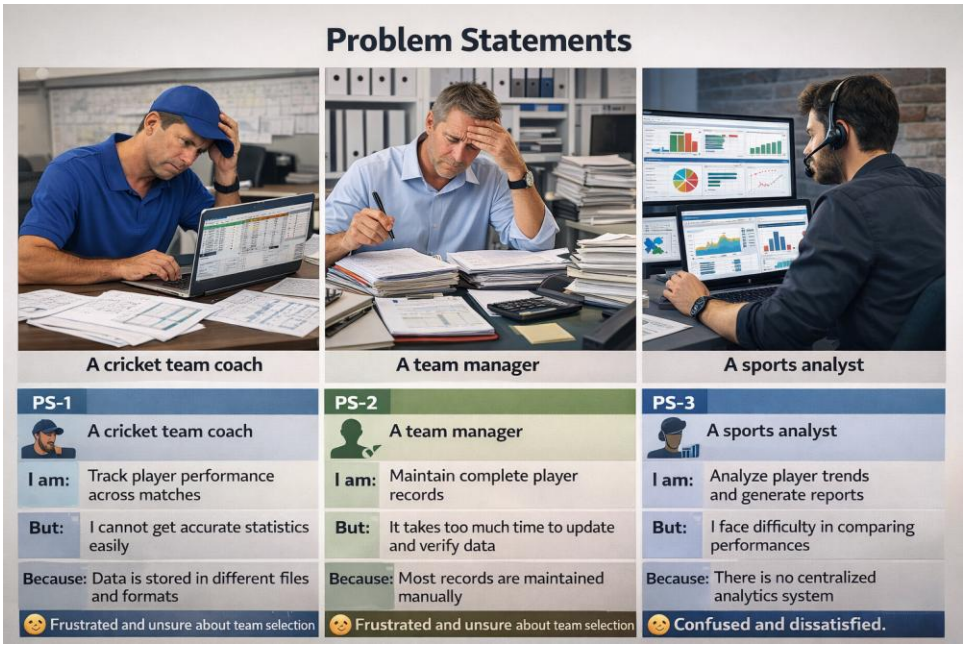
The main purpose of this project is:

- To digitize player performance records
- To improve accuracy and accessibility of cricket statistics
- To provide real-time analytics
- To support player development and team planning
- To automate data management using Salesforce

2. Ideation Phase

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A cricket team coach	Track player performance across matches	I cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated and unsure about team selection
PS-2	A team manager	Maintain complete player records	It takes too much time to update and verify data	Most records are maintained manually	Stressed and inefficient
PS-3	A sports analyst	Analyze player trends and generate reports	I face difficulty in comparing performances	There is no centralized analytics system	Confused and dissatisfied



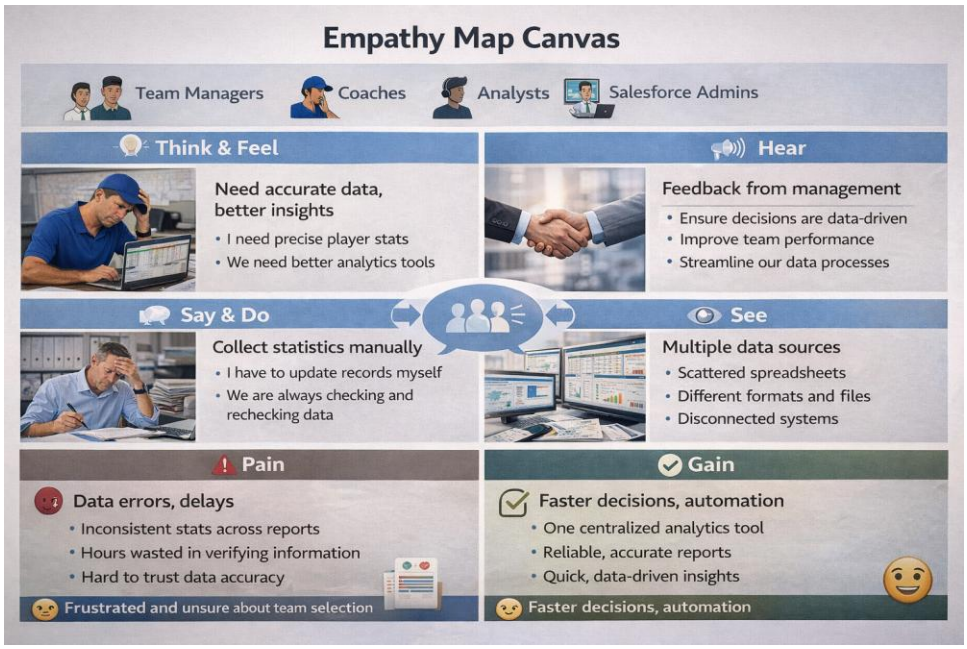
2.2 Empathy Map Canvas

Target Users

- Team Managers
- Coaches
- Analysts
- Salesforce Admins

Empathy Analysis

- Think & Feel: Need accurate data, better insights
- Hear: Feedback from management
- See: Multiple data sources
- Say & Do: Collect statistics manually
- Pain: Data errors, delays
- Gain: Faster decisions, automation

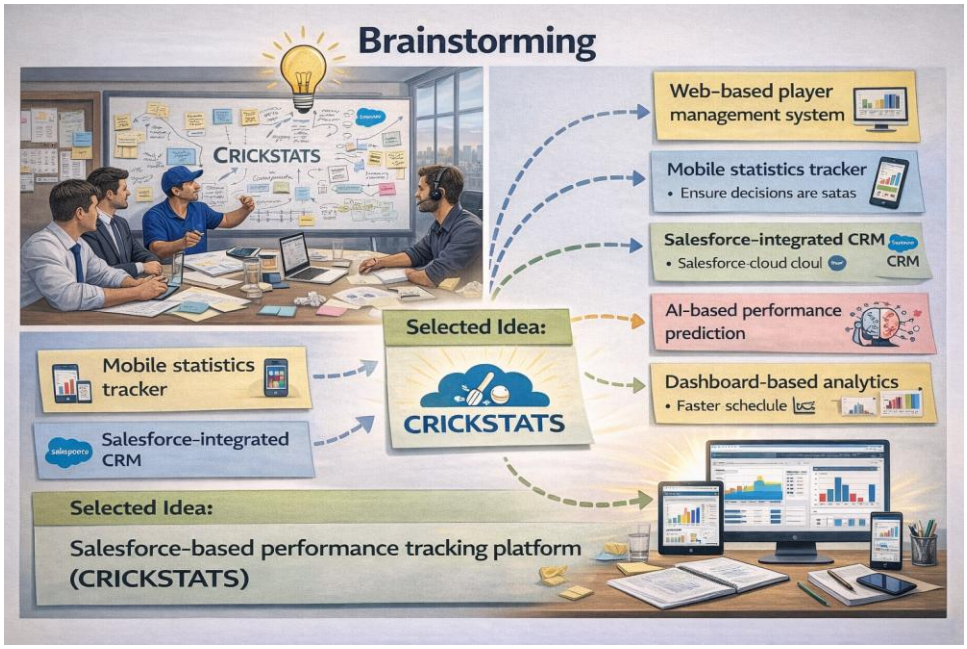


2.3 Brainstorming

Brainstormed Ideas

- Web-based player management system
- Mobile statistics tracker
- Salesforce-integrated CRM
- AI-based performance prediction
- Dashboard-based analytics

Selected Idea: Salesforce-based performance tracking platform (CRICKSTATS)



3. Requirement Analysis Phase

3.1 Customer Journey Map

Journey Stages

- Login
- Enter Player Data
- Record Match Performance
- Generate Reports
- Analyze Results
- Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Player Registration
FR-2	Match Performance Entry
FR-3	Dashboard Generation
FR-4	Report Creation
FR-5	Flow Automation

Non-Functional Requirements (NFRs) — CRICKSTATS

NFR-1: Usability

The system should be easy to understand and use by coaches, managers, analysts, and admins without requiring technical training.

- Simple and intuitive user interface
- Clear navigation menus
- Easy-to-read dashboards
- Minimal learning curve

Example: A new coach should be able to log in, enter match data, and view player reports within a few minutes without special training. The Salesforce Lightning interface provides drag-and-drop dashboards and user-friendly forms.

NFR-2: Security

The system must protect sensitive player data from unauthorized access.

- Role-based access control
- Secure authentication
- Encrypted data storage
- Regular backups

Example: Coaches can view player statistics; managers can edit records; analysts can generate reports; only admins can modify system settings. Salesforce Profiles and Permission Sets restrict access to confidential information.

NFR-3: Reliability

The system should function correctly without failure and store data accurately.

- No data loss
- Automatic backups
- Error handling
- System stability

Example: If a manager enters match statistics and loses internet connection, the data should be saved automatically when reconnected. Salesforce cloud backup ensures recovery in case of system failure.

NFR-4: Performance

The system should respond quickly and handle large volumes of data efficiently.

- Fast page loading
- Quick report generation
- Smooth dashboard rendering
- Minimal response time

Example: When an analyst generates a performance report for 100+ players, the report should load within 2–3 seconds. Optimized Salesforce queries and caching improve response speed.

NFR-5: Availability

The system should be accessible whenever users need it.

- 24/7 system availability
- Minimal downtime
- Cloud-based access
- Disaster recovery

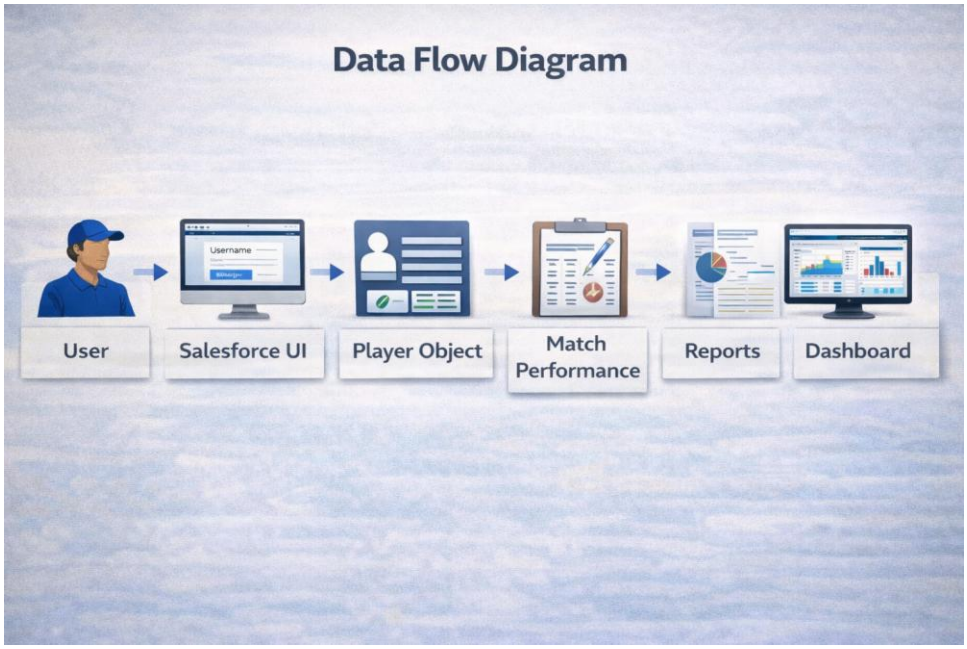
Example: Coaches should be able to access CRICKSTATS even during tournaments at night or on weekends. Salesforce provides high uptime with cloud hosting.

NFR-6: Scalability

The system should support increasing numbers of users, players, and records.

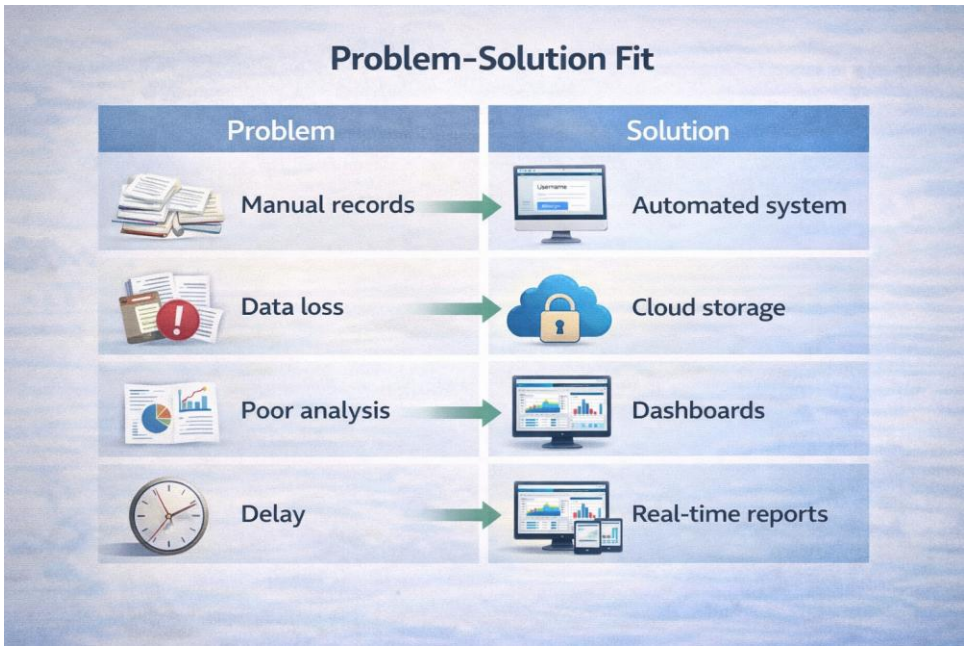
- Handle growing datasets
- Support more teams and leagues
- Expand storage capacity
- Maintain performance under load

Example: If CRICKSTATS is extended from one team to 50 teams, the system should still perform smoothly. Salesforce cloud infrastructure automatically scales storage and processing.



3.3 Data Flow Diagram

Data Flow: User → Salesforce UI → Player Object → Match Performance → Reports → Dashboard



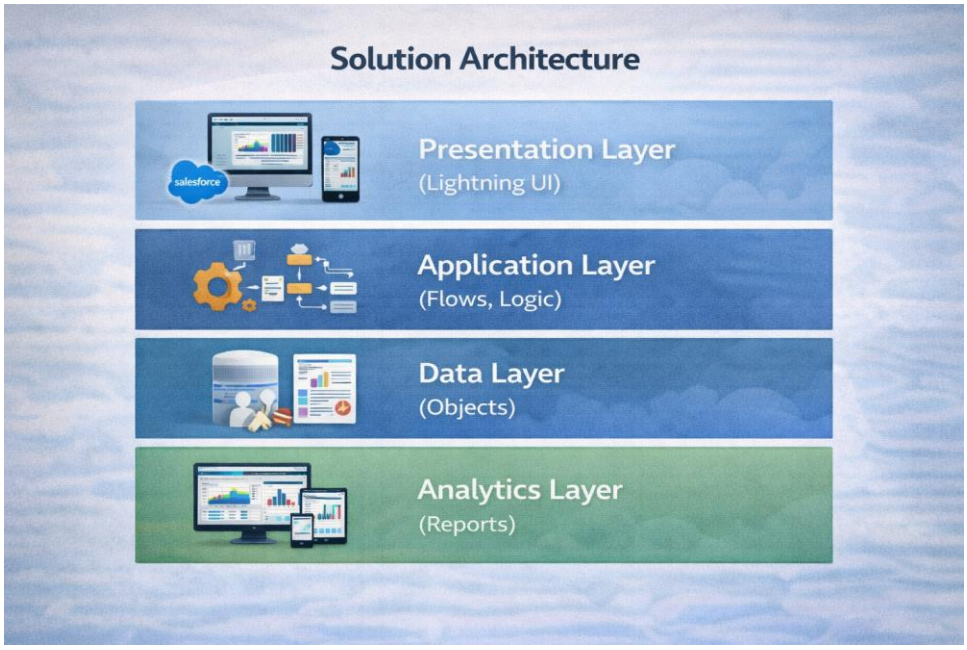
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning
Analytics	Reports & Dashboards
Automation	Salesforce Flows
Backend	Salesforce Objects
Database	Salesforce Cloud

4. Project Design Phase

4.1 Problem–Solution Fit

The CRICKSTATS system effectively addresses the key challenges faced in traditional cricket performance management. Earlier, most player and match data were maintained using manual records, which were time-consuming and prone to errors. This issue has been resolved by implementing an automated system through Salesforce, which ensures accurate and efficient data entry and management. The problem of data loss, caused by improper storage and lack of backups, is overcome by using secure cloud storage that safely stores all information and allows easy retrieval. Poor analysis of player performance, due to scattered and unorganized data, is addressed through interactive dashboards that present clear visual insights and reports. Additionally, delays in accessing performance information and generating reports are eliminated by providing real-time reporting features, enabling users to make faster and more informed decisions. Thus, CRICKSTATS successfully bridges the gap between existing problems and modern technological solutions.



4.2 Proposed Solution

CRICKSTATS provides:

- Custom objects (Player, Match Performance)

- Master-detail relationships
- Roll-up summaries
- Lightning App
- Automated flows
- Interactive dashboards

Key Features

- Centralized database
- Live statistics
- Secure access
- Scalable design
- Easy reporting

4.3 Solution Architecture

Architecture Layers

- Presentation Layer (Lightning UI)
- Application Layer (Flows, Logic)
- Data Layer (Objects)
- Analytics Layer (Reports)

5. Project Planning & Scheduling

5.1 Project Planning

- Agile-based execution
- Object creation → automation → reporting
- Sprint-wise development

Product Backlog, Sprint Schedule, and Estimation

Sprint	Epic	USN	User Story / Task	Pts	Priority	Owner
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the CRICKSTATS system.	1	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Player and Match Performance objects so that player and match data can be stored systematically.	3	High	Member2
Sprint-2	Navigation	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage records within the application.	3	High	Member3
Sprint-2	UI Design	USN-4	As a coach, I want a dedicated Lightning App so that all performance	3	High	Member4

Sprint	Epic	USN	User Story / Task	Pts	Priority	Owner
			tracking features are accessible from one dashboard.			
Sprint-2	Data Structure	USN-5	As an admin, I want to create player and match fields so that complete performance details can be captured accurately.	5	Medium	Member1
Sprint-2	Data Structure	USN-6	As a user, I want customized page layouts so that only relevant fields are displayed during data entry.	5	High	Member2
Sprint-2	Automation	USN-7	As a system, I want to validate player and match inputs so that incorrect data entries are prevented.	3	High	Member4
Sprint-3	Automation	USN-8	As a system, I want automated flows to update player statistics so that performance data remains accurate.	5	High	Member1

Project Tracker, Velocity & Burndown Chart

Sprint	Total Pts	Duration	Start Date	End Date (Planned)	Pts Completed	Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase

Project Flow

- Milestone 1: Creation of developer account
- Milestone 2: Object Creation
- Milestone 3: Tabs
- Milestone 4: The Lightning App
- Milestone 5: Fields & Relationships
- Milestone 6: Roll-up Summary
- Milestone 7: Flows

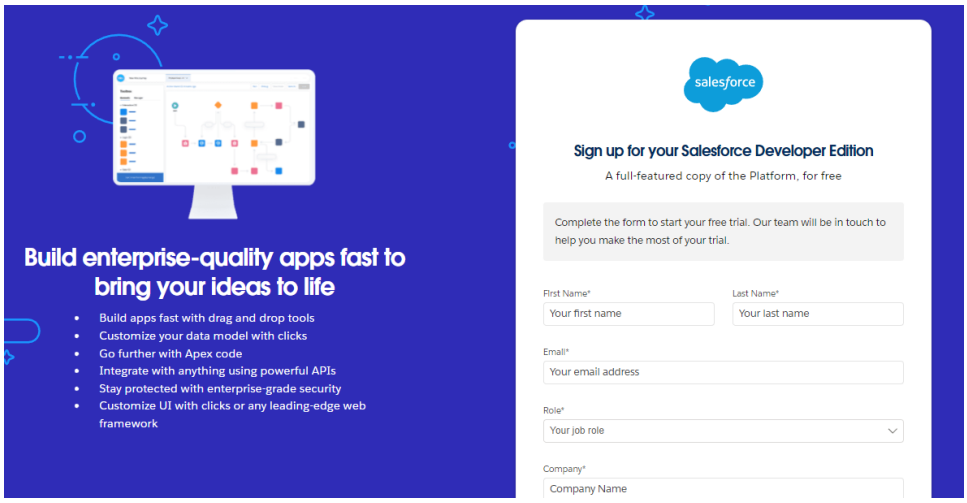
Milestone 1 — Salesforce Developer Account Creation

Activity 1: Creating Developer Account

Creating a developer org in Salesforce.

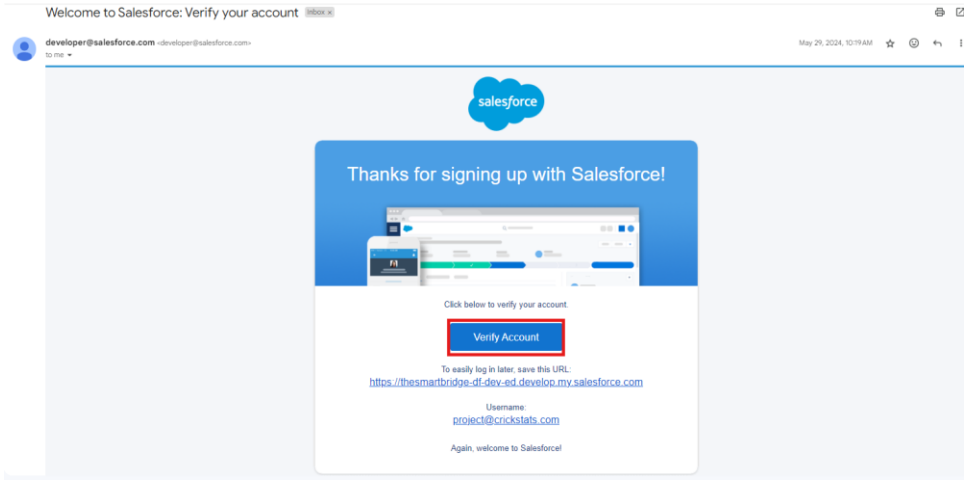
1. Go to https://developer.salesforce.com/signup
2. On the sign-up form, enter the following details:
    - First name & Last name
    - Email
    - Role: Developer
    - Company: College or Company Name
    - Country: India
    - Postal Code: pin code
    - Username: should be a combination of your name and company

This need not be an actual email ID — you can give anything in the format username@organization.com. Click Sign Me Up after filling these.

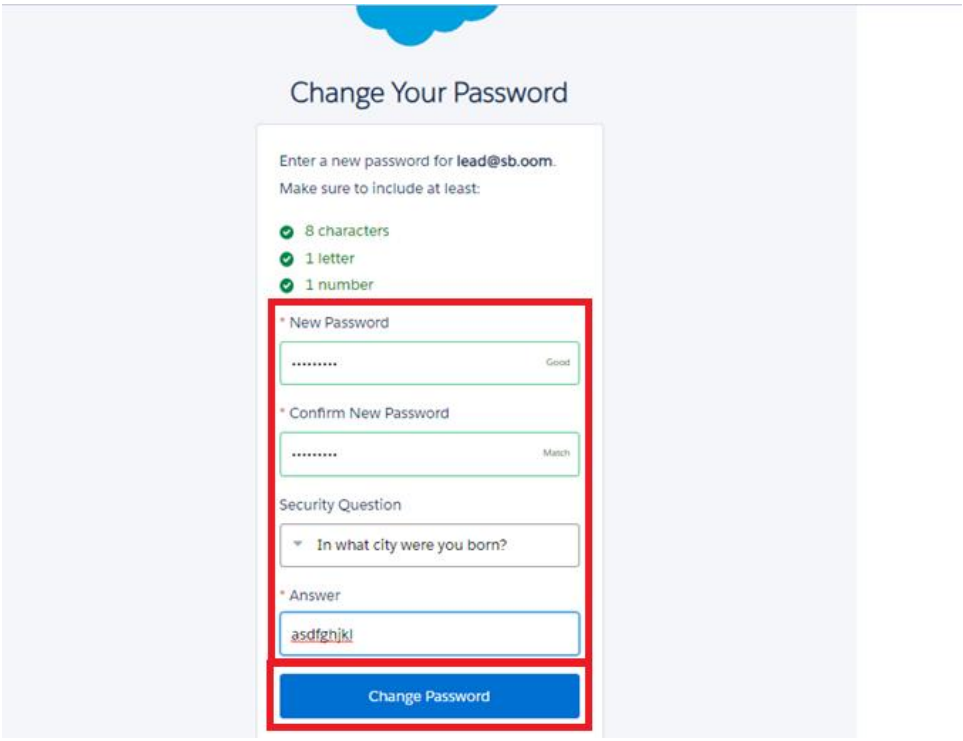


Activity 2: Account Activation

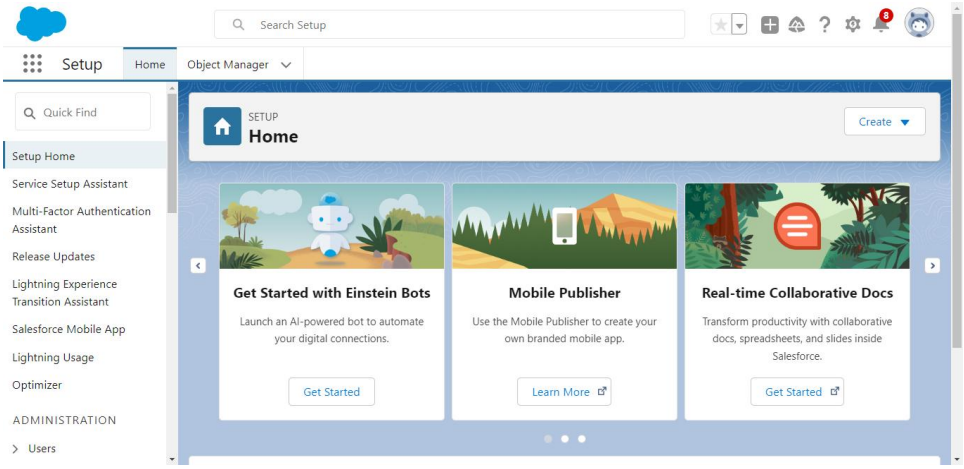
1. Go to the inbox of the email used while signing up and click Verify Account to activate the account (may take 5–10 minutes).
2. Click Verify Account.



1. Give a password, answer a security question, and click Change Password.



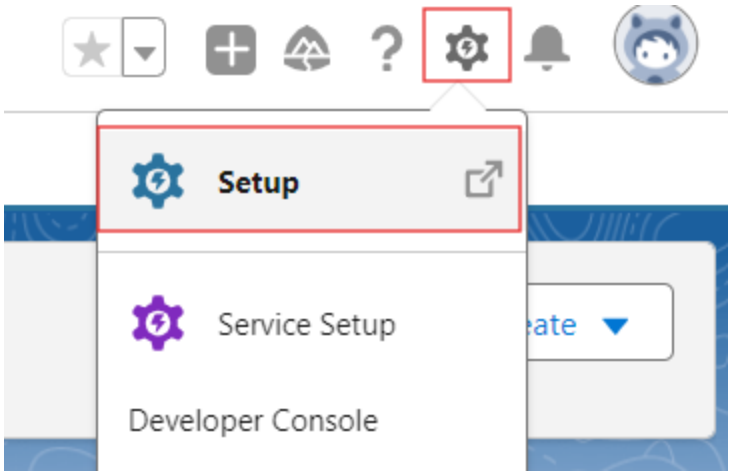
2. You will be redirected to the Salesforce Setup page.



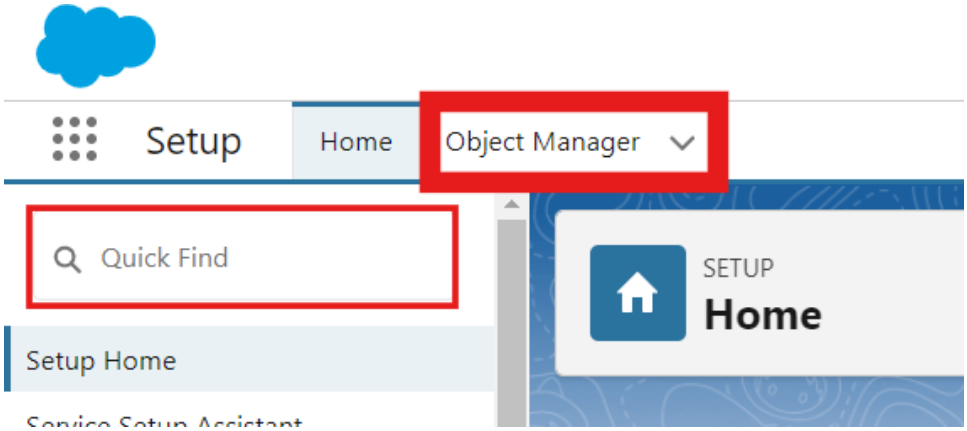
Milestone 2 — Object Creation

Activity 1: Player Object

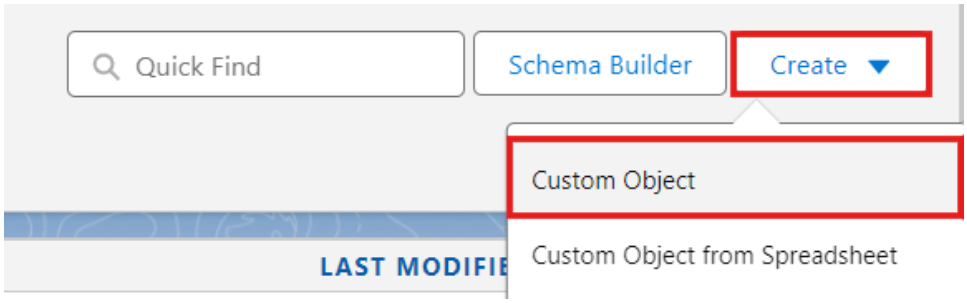
1. In your Salesforce org, click the gear icon (top left) and select Setup.



2. Click the Object Manager tab (or enter it in the Quick Find box).



3. On the Object Manager page, click Create | Custom Object.



From the setup page → Object Manager → Create → Custom Object:

- Label: Player
- Plural Label: Players
- Record Name: Player Name (Data Type: Text)
- Optional features: check Allow Reports | Allow Activities | Track Field History
- Search Status: check Allow Search
- Save

Activity 2: Match Performance Object

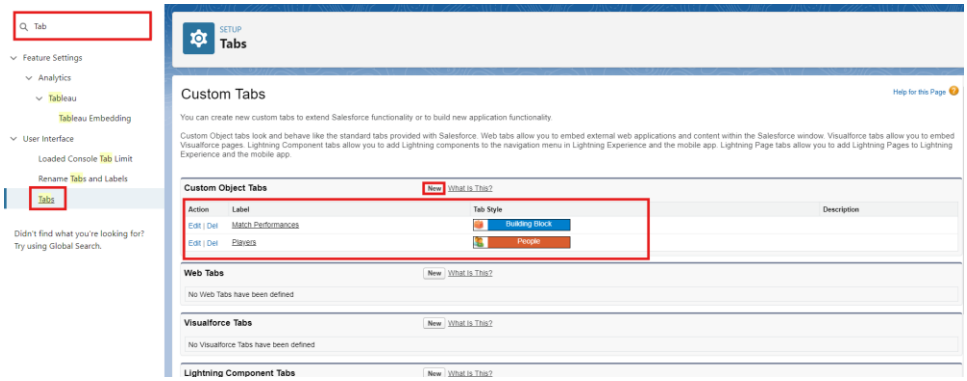
From the setup page → Object Manager → Create → Custom Object:

- Label: Match Performance
- Plural Label: Match Performances
- Record Name: Customer Account Name (Data Type: Text)
- Optional features: check Allow Reports | Allow Activities | Track Field History
- Search Status: check Allow Search
- Save

Milestone 3 — Tabs

Activity 1: Custom Tab Creation

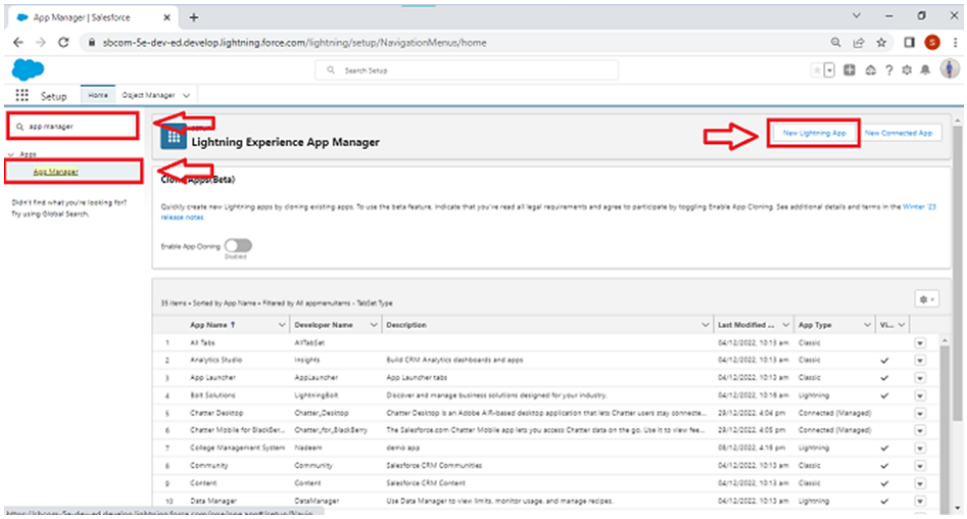
1. Go to Setup → type Tabs in Quick Find → click Tabs → New (under Custom Object Tab).
2. Select Object (Player) → select the tab style → Next (Add to Profiles, keep default) → Next (Add to Custom App, uncheck Include Tab).
3. Make sure Append tab to users' existing personal customizations is checked.
4. Click Save.
5. Repeat the same steps for Match Performances.



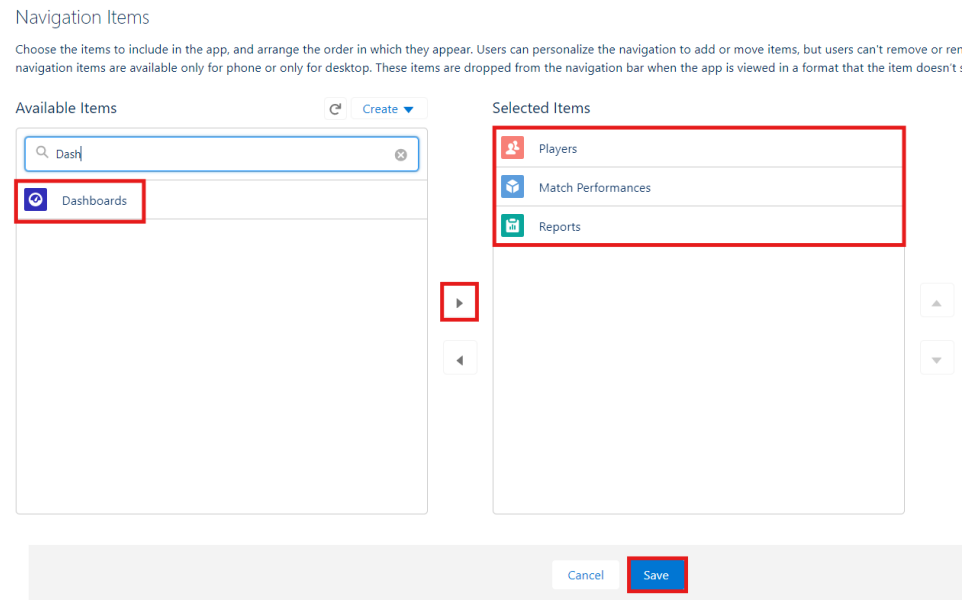
Milestone 4 — The Lightning App

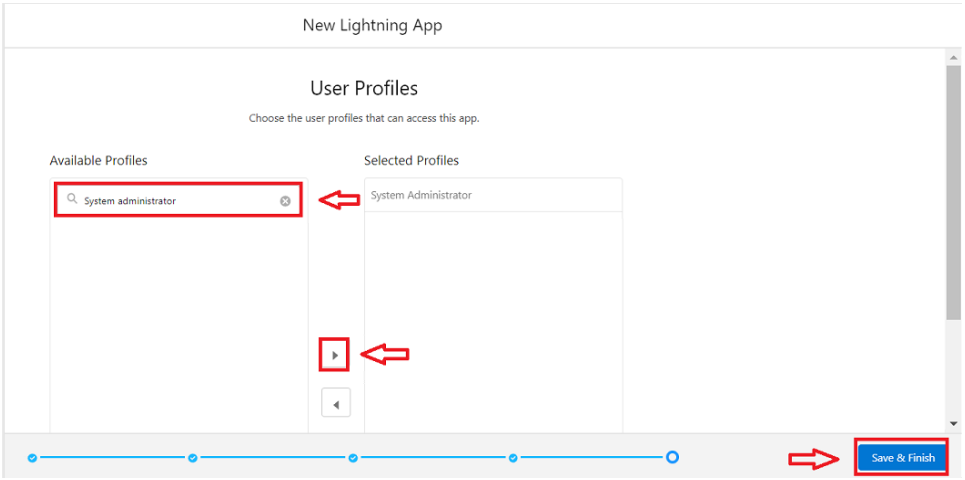
Activity 1: Create a Lightning App

1. Go to Setup → search "app manager" in Quick Find → select App Manager → click New Lightning App.

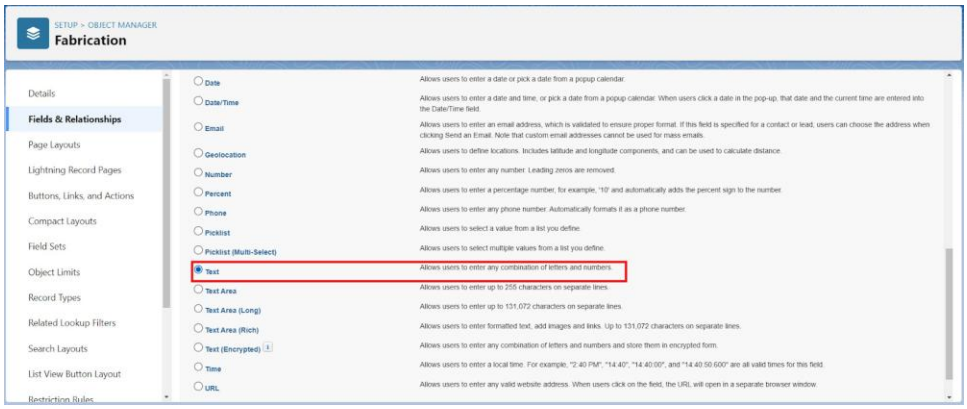


2. Fill in app details and branding: App Name — CRICKSTATS; Developer Name auto-populates; add an image (optional) and primary color hex, or leave default. Click Next.
3. In App Options: Navigation style — Standard navigation; Setup experience — setup; Supported form factors — Desktop and phone. Click Next.
4. In Utility Items: Utility Bar alignment — Default. Click Next.
5. In Navigation Items, select the created custom objects and required standard objects: Players, Match Performances, Reports, Dashboards.





6. To add User Profiles: System Administrator.

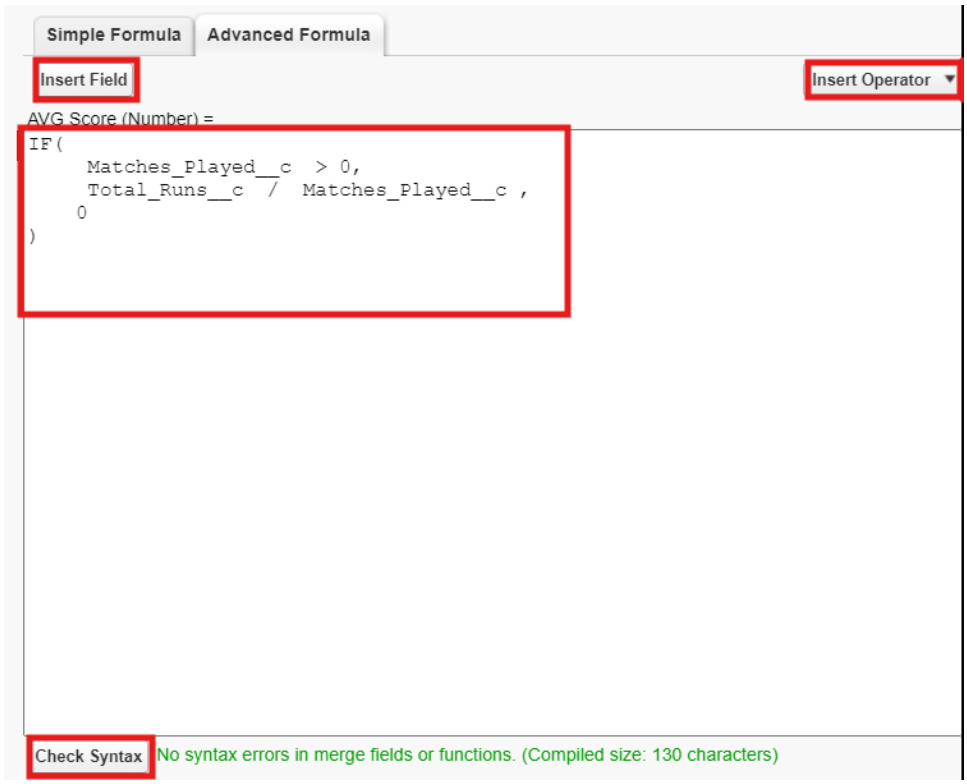


7. Click Save & Finish.

Milestone 7 — Flows

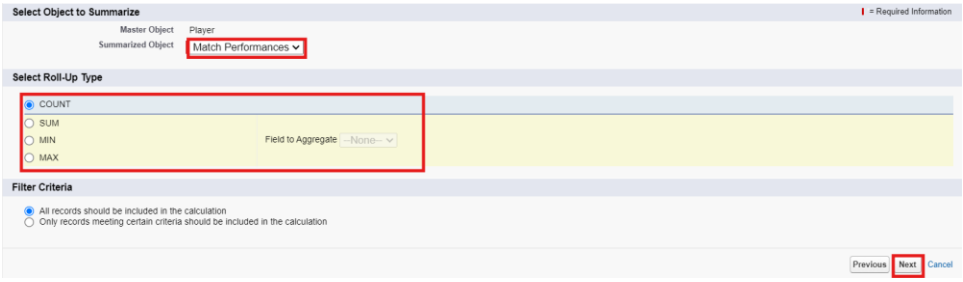
Activity 1: Create Custom Fields for Player Object

From Object Manager → Player → Fields & Relationships → New, the following custom fields were created:



- Player Name — Text (50)
- Player Number — Text (3)
- Profile Picture — Rich Text Area
- Date of Birth — Date
- Role — Picklist: Batsman, Bowler, Batsman & Wicket Keeper, All Rounder
- Batting Style — Picklist: Right-Handed, Left-Handed

- Bowling Style — Picklist: Right-Arm Fast, Left-Arm Fast, Right-Arm Medium, Left-Arm Medium, Right-Arm Spin, Left-Arm Spin
- Nationality — Picklist: Australia, Bangladesh, England, India, New Zealand, Pakistan, South Africa, Sri Lanka, West Indies, Afghanistan, Zimbabwe, Ireland, Scotland, Netherlands, United Arab Emirates, Nepal, United States of America
- Team — Picklist: Chennai Super Kings (CSK), Delhi Capitals (DC), Kolkata Knight Riders (KKR), Mumbai Indians (MI), Punjab Kings (PBKS), Rajasthan Royals (RR), Royal Challengers Bangalore (RCB), Sunrisers Hyderabad (SRH), Gujarat Titans (GT), Lucknow Super Giants (LSG)
- AVG Score — Formula (Number): IF(Matches_Played__c > 0, Total_Runs__c / Matches_Played__c, 0)
- Match Performance — Lookup Relationship, related to Match Performance



Activity 2: Create Custom Fields for Match Performance Object

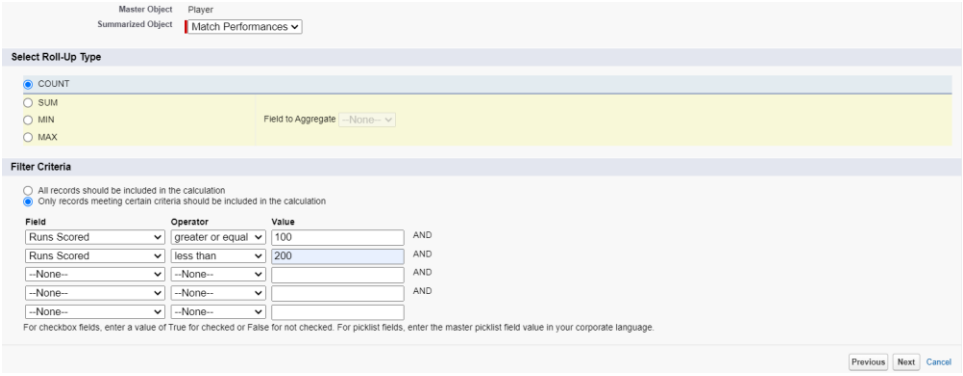
- Match Performance Name — Text (80)
- Player of the Match — Checkbox
- Match Result — Picklist: WON, LOST, DRAW
- Runs Scored — Number
- Balls Faced — Number
- Fours — Number
- Sixes — Number
- Overs Bowled — Number
- Wickets Taken — Number
- Runs Conceded — Number
- Maiden Overs — Number
- Stumpings — Number
- Catches — Number
- Player — Master-Detail Relationship, related to Player

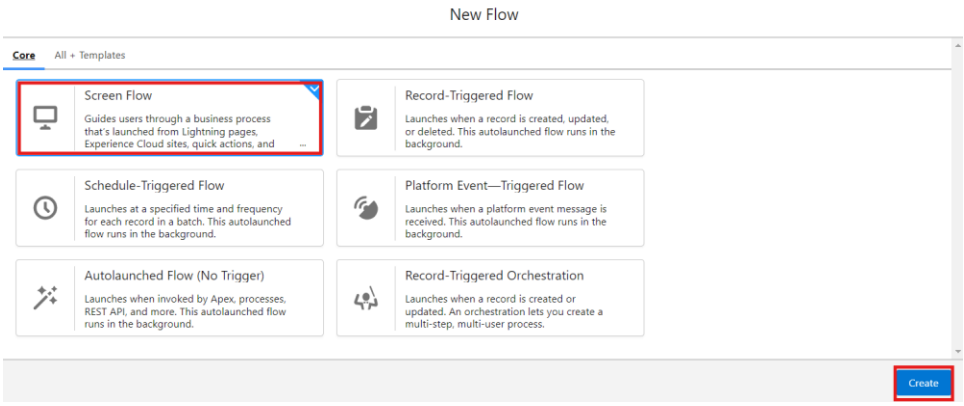
Milestone 6 — Roll-up Summary

Note: Roll-up Summary fields can only be created on the parent object in a Master-Detail Relationship. The following roll-up summary fields were created on the Player object, summarizing Match Performances:

- Matches Played — COUNT, all records
- Number of 50's — COUNT, filtered where Runs Scored ≥ 50 and < 100
- Number of 100's — COUNT, filtered where Runs Scored ≥ 100 and < 200
- Total 4's — SUM of Fours, all records
- Total 6's — SUM of Sixes, all records
- Total Runs — SUM of Runs Scored, all records

- Total Overs — SUM of Overs Bowled, all records
- Total Wickets — SUM of Wickets Taken, all records
- Total Stumpings — SUM of Stumpings, all records
- Total Catches — SUM of Catches, all records



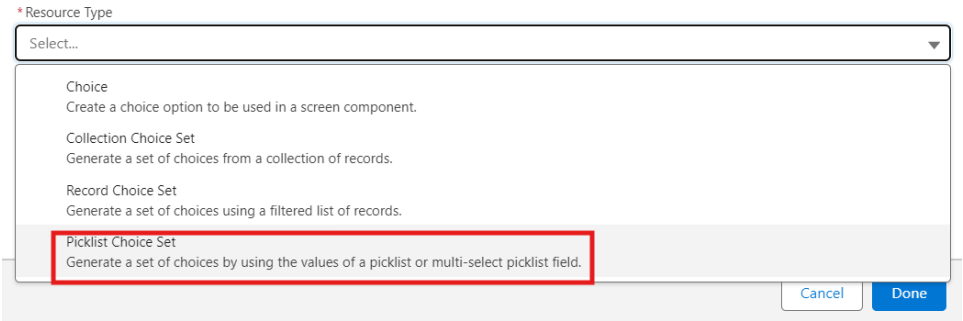


Milestone 7 — Flows

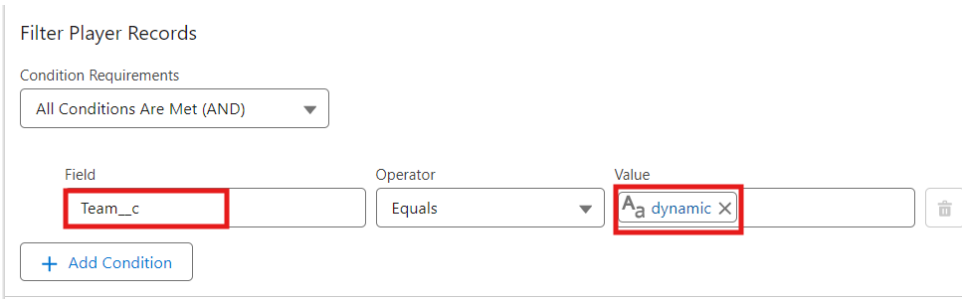
Activity 1: Screen Flow

A Screen Flow named Player Flow (API: Player_Flow) was built to let a user pick a team and view that team's player statistics.

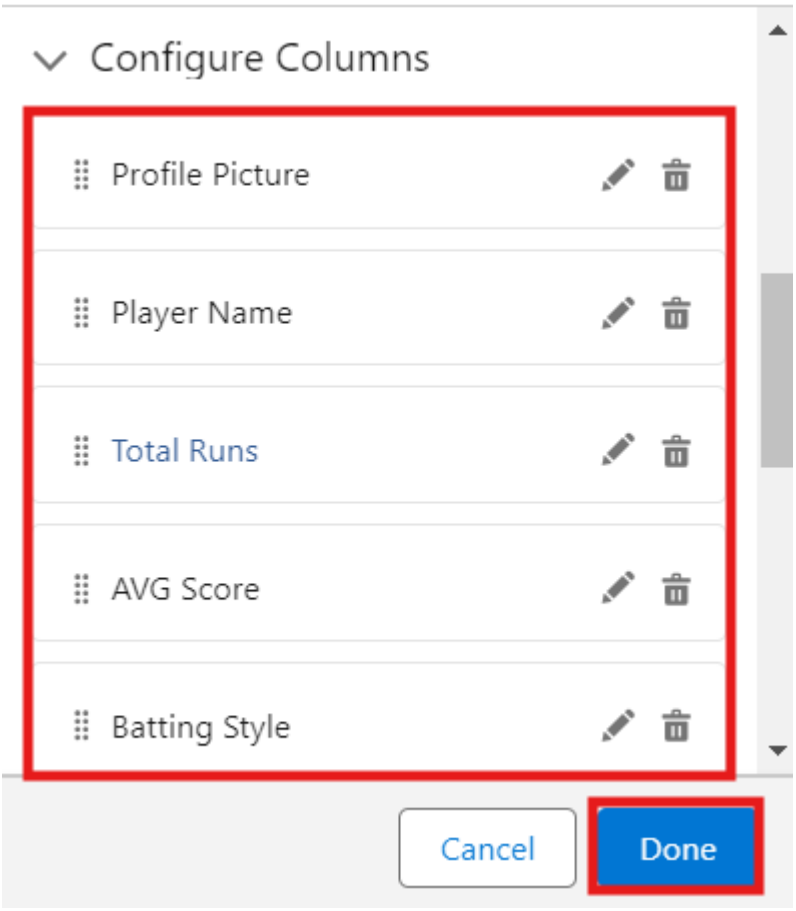
1. Go to Setup, type Flow in Quick Find → click Flows → New Flow → select Screen Flow → Create.



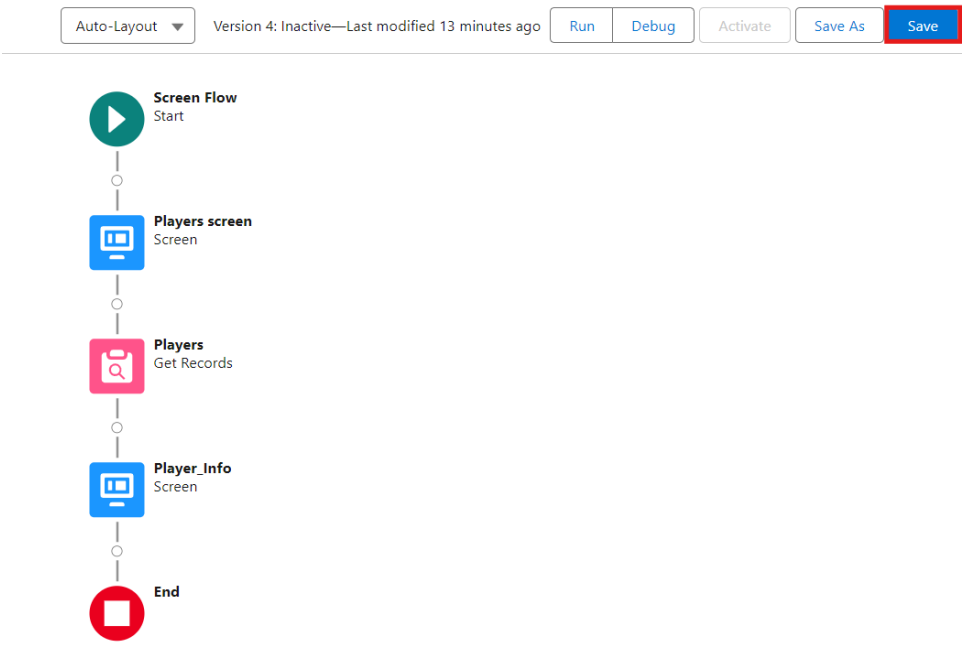
2. Add a Screen component labeled Players_screen. Place a Picklist component labeled Team (API: Team), Component Type: Picklist, Data Type: Text, and create a new Picklist Choice Set resource (API: Team; Object: Player; Field: Team).



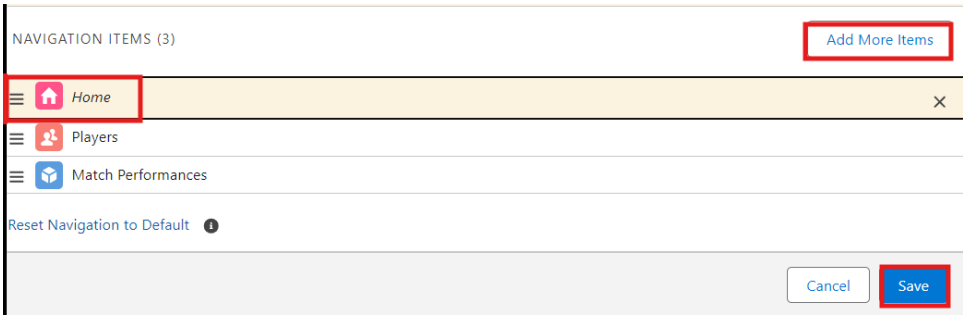
3. Add a Get Records component labeled Players, filtering Player records where Team equals a new Text variable resource (API: Dynamic) whose default value is the Team picklist screen component.



4. Add a second Screen component labeled Player_info, containing a Data Table (API: Player_info_table) sourced from the Players collection, with columns: Profile Picture, Player Name, Total Runs, AVG Score, Batting Style, Bowling Style, Matches Played, Number of 50's, Number of 100's.

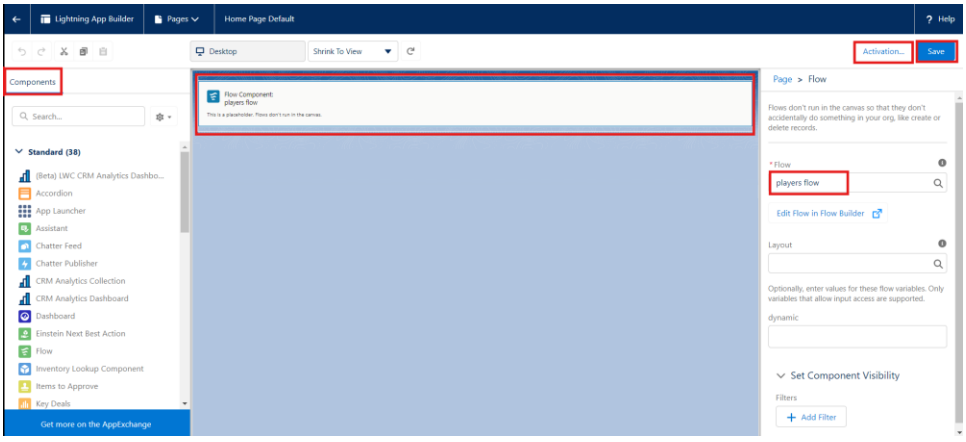


5. Save the flow as Player Flow (API: Player_Flow).

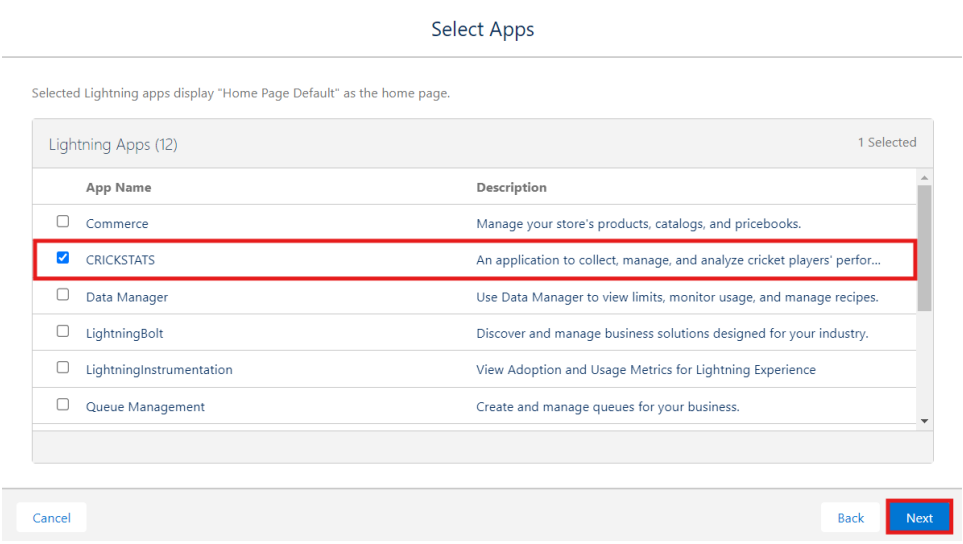


Activity 2: Flow on Home Page

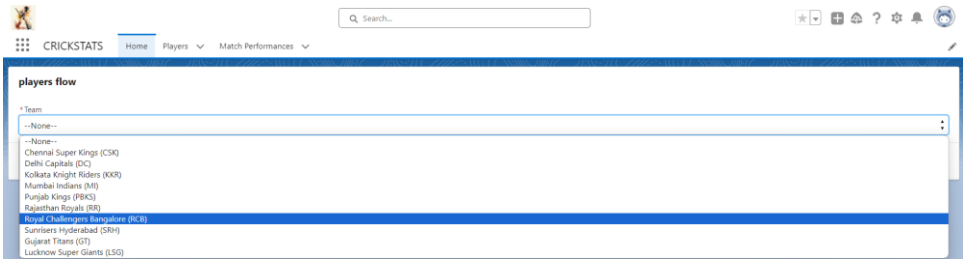
1. On the CRICKSTATS app, click the Personalize your nav bar icon (pencil), add a Home page, and Save.



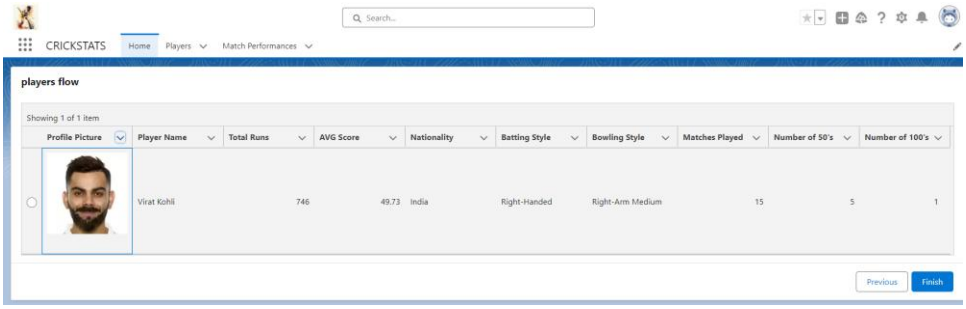
2. Remove all other components on the Home page and add the Flow component; set Flow to Players Flow.



3. Click Save and Activation.
4. Click App Default and select Assign to Apps.



5. Select the CRICKSTATS app, click Next, then Save.



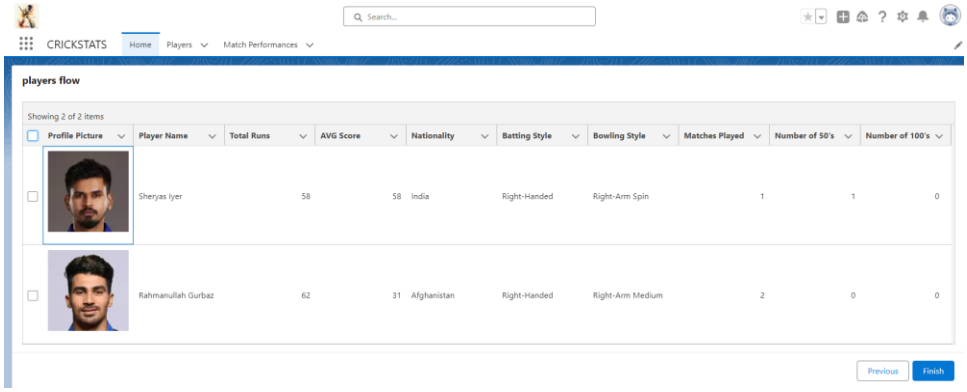
7. Functional & Performance Testing

During the testing phase, screenshots were captured from the Salesforce Lightning user interface to verify the correct functioning of all implemented features in the CRICKSTATS system. These include record creation for Player and Match Performance objects; validation rule checks for mandatory fields, score limits, and input constraints; verification of record type behavior for Domestic and International match categories; and flow execution for automatic updating of player statistics and data consistency. The functionality of roll-up summary fields was also tested to ensure accurate calculation of total runs, wickets, averages, and performance metrics.

In addition, system performance was evaluated by monitoring response time during data entry, report generation, and dashboard loading. The application was tested under multiple user interactions to ensure stable performance without delays. Screenshots of generated reports and interactive dashboards were documented to

confirm accurate data visualization, real-time updates, and reliable performance tracking. These testing activities ensured that the CRICKSTATS platform met both functional and performance requirements and was ready for deployment.

Note: Records were entered in the Player and Match Performance objects to generate the results below.



8. Advantages & Disadvantages

Advantages

- Centralized system
- Real-time analytics
- Secure platform
- Scalable design
- Reduced manual work

Disadvantages

- Requires Salesforce license
- Internet dependency
- Limited offline access

9. Conclusion

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making.

The project demonstrates practical application of SDLC, Agile methodology, and CRM-based development.

10. Future Scope

- Mobile application integration
- AI-based performance prediction
- Video analysis integration
- Player ranking system

- Public statistics portal
- Multi-league support

11. Appendix

- Salesforce Objects & Fields
- Manual match statistics
- Flow Configurations

Thank You